

py-multiproc

python multiprocessing ML project example

2024-10-09 Wednesday

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1 PROJECT

1.1 py-multi Notebooks

- Interprocess Queue Performance
- Interprocess Pickle Performance
- (,) Artur Glogowski (2020) Red Hat Security: Identity Management and Active Directory Integration - v7.4, Red Hat.

2 Multiprocess Architecture

2.1 MP Components

2.1.1 MP Components Overview

```

RLModelSender -l- RTModelReceiver: ;RL-Models;
RTController -r- RLTrainer: ;RT-Ticks;
RTEpisodePublisher -r- RLEpisodeReader: ;RT-Episodes;
PXCLIRequestor -r- PXInit: ;CLI-Commands;

```

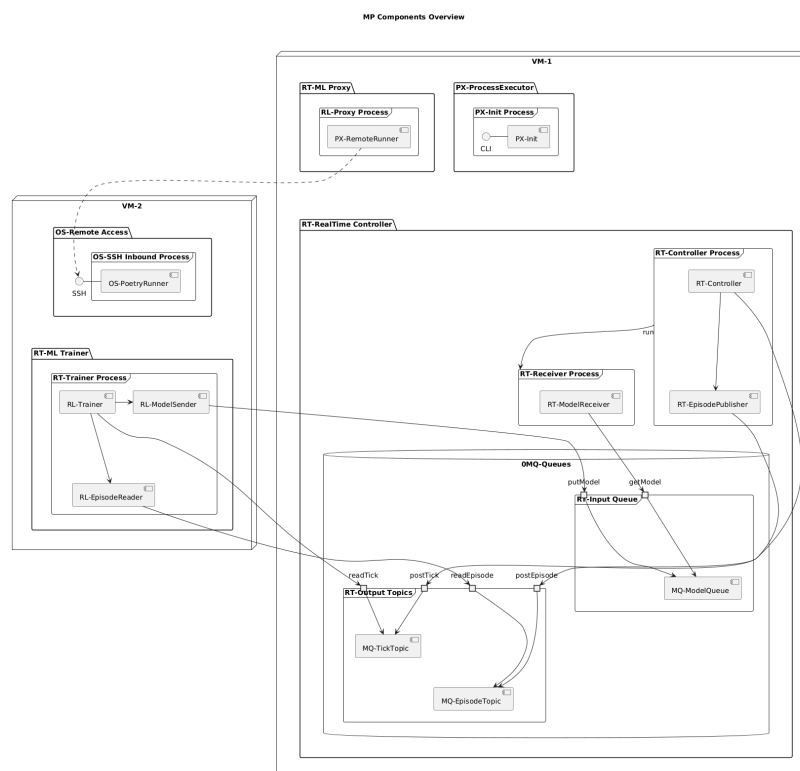
```

"PX-Init Main Process" -d- "RT-Controller Process": run
"PX-Remote Launcher Process" -r- "OS-SSH Inbound Process": run
"OS-SSH Inbound Process" -d- "RT-Trainer Process": run
"PX-Init Main Process" -r- "RL-Proxy Process": run

```

PXRemoteRunner .r. OSSSHService

note left of PXCLIRequestor: optional CLI support



```

##
#
#

```

UP=1; apt update && apt upgrade; apt autoremove -y

date

3 REFERENCES

3.1 Notes

- 30multi

3.2 Library Documentation

- (Python and Stdlib, 2024) Python & Stdlib (2024) multiprocessing — Process-based parallelism, .
- (ZeroMQ, 2023) ZeroMQ (2023) ØMQ - The Guide, .
- (ZeroMQ, 2023a) ZeroMQ (2023) PyZMQ Documentation, .
- (ZeroMQ, 2023a) ZeroMQ (2023) Using PyZMQ, .

3.3 Blog Articles

- (Moraneus, 2024) Moraneus (2024) Understanding Multithreading and Multiprocessing in Python, .
- (Zhe Ming Chng, 2022) Zhe Ming Chng (2022) A Gentle Introduction to Serialization for Python, .

4 Refs

4.1 org-refs

`printbibliography:`

4.2 bib-refs

References

- [1] Python and Stdlib. (2024) multiprocessing — process-based parallelism. [Online]. Available: <https://docs.python.org/3/library/multiprocessing.html>
- [2] ZeroMQ. (2023) Ømq - the guide. [Online]. Available: <https://zguide.zeromq.org/>
- [3] —. (2023) Pyzmq documentation. [Online]. Available: <https://pyzmq.readthedocs.io/en/latest/#>

- [4] ——. (2023) Using pyzmq. [Online]. Available: <https://pyzmq.readthedocs.io/en/latest/howto/index.html>
- [5] Z. M. Chng. (2022) A gentle introduction to serialization for python. [Online]. Available: <https://machinelearningmastery.com/a-gentle-introduction-to-serialization-for-python/>
- [6] Moraneus. (2024) Understanding multithreading and multiprocessing in python. [Online]. Available: <https://medium.com/@moraneus/understanding-multithreading-and-multiprocessing-in-python-1ed39bb078d5>